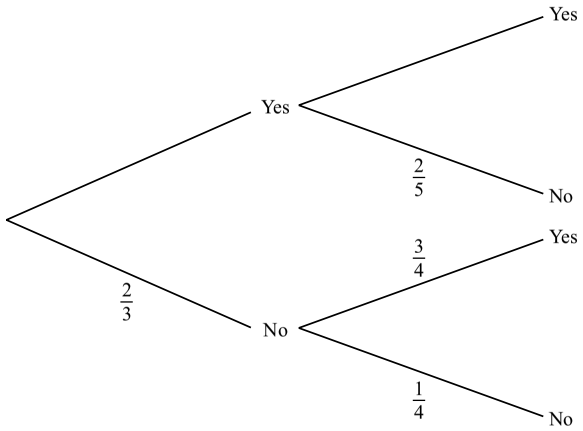
| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 1 | –2, –1, 0, 1 | **2** | **B1** if one error/omission/extra |
| 2 | Correct triangle with arcs | **2** | **B1** correct no arcs |
| 3 | $x^9y^4$ final answer | **2** | **B1** for $x^9y^k$ or $x^ky^4$ or correct answer seen then spoiled |
| 4 | –14, –5, 1 | **2** | **B1** for 2 correct |
| 5(a) | 32.7 to 33.3 | **1** | **FT** *their* printed diagram |
| 5(b) | 118 to 120 | **1** | **FT** *their* printed diagram |
| 5(c) | 298 to 300 | **2** | **B1** for 360 – (180 – *their* **(b)**)<br>**FT** *their* **(b)** + 180 |
| 6(a) | 5 | **2** | **M1** for 10 × 12 × 'h' = 600 |
| 6(b) | 172 or 172.4 to 172.5 | **2** | **M1** for $1120 \times \frac{4}{3} \times \pi \times 0.45^3$ |
| 7(a) | 20 | **1** | |
| 7(b) |  <br>$\frac{2}{3}, \frac{2}{5}, \frac{3}{4}, \frac{1}{4}$ oe | **2** | **M1** for 2 or 3 correct |
| 7(c) | $\frac{7}{10}$ oe | **3** | **M2** for $\frac{1}{3} \times \frac{3}{5}$ + *their* $\frac{2}{3}$ × *their* $\frac{3}{4}$<br>or **M1** for one product |

© Cambridge University Press & Assessment 2022

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 8(a) | $y = x$ ruled<br>$x + y = 7$ ruled | 3 | **B1** for $y = x$<br>**B2** for $x + y = 7$<br>or **B1** for ruled line with gradient –1 or ruled line through (0, 7) or (7, 0) but not $x = 7$ or $y = 7$. |
| 8(b) | Correct region below $y = x$, above $x + y = 7$ and above $x$-axis labelled R | 1 | |
| 9(a) | $\frac{8}{3}$ oe | 1 | |
| 9(b) | 92 | 2 | **M1** for correct calculation of an area |
| 10 | $(x + 3)(2 – 3y)$ | 2 | **M1** for $2(x + 3) – 3y(x + 3)$<br>or $x(2y – 3) + 3(2y – 3)$ |
| 11(a) | 6 | 2 | **M1** for $10 + 13 – (20 – 3)$ oe or correctly completed Venn diagram |
| 11(b) | 7 | 1 | **FT** *their* Venn diagram or 13 – *their* **(a)** |
| 12 | 160.375 | 4 | **M1** for mid-values for three of 125, 155, 162.5, 175<br>**M1** for $\Sigma fx$ where $x$ is in correct interval including boundary<br>**M1 dep** $\frac{\Sigma fx}{100}$ dep on second **M1** |
| 13(a) | 49.4 or 49.39 to 49.40 | 2 | **M1** for $\tan = \frac{14}{12}$ oe |
| 13(b) | 22.4 or 22.36 to 22.37 | 4 | **B1** for 62 + *their* **(a)**<br>**M2** for $\sqrt{12^2 + 15^2 - 2 \times 12 \times 15 \cos(62 + \textit{their } \mathbf{(a)})}$<br>OR<br>**B1** for 62 + *their* **(a)**<br>**M1** for<br>$12^2 + 15^2 – 2 \times 12 \times 15 \cos(62 + \textit{their } \mathbf{(a)})$<br>**A1** for 500.3 to 500.4 |
| 13(c) | 13.2 or 13.24... | 3 | **M2** for $\sin 62 = \frac{\text{dist}}{15}$<br>or **M1** for recognition of shortest distance soi |

© Cambridge University Press & Assessment 2022

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 19 | 391 or 391.1 to 391.2 | 4 | **M3** for $3 \times 6 \times 20 + 2 \times \frac{1}{2} \times 6 \times 6 \times \sin 60$ oe<br>OR<br>**M1** for [3 ×] $6 \times 20$<br>**M1** for [2 ×] $\frac{1}{2} \times 6 \times 6 \times \sin 60$ oe |
| 20 | [$k$ =] 9<br>[$u$ =] 3 | 2 | **B1** each |
| 21 | $\frac{5p}{2(p+4)}$ or $\frac{5p}{2p+8}$ final answer | 3 | **B1** for $5p(p-4)$<br>**B1** for $2(p+4)(p-4)$ |
| 22(a)(i) | $-\frac{1}{3}\mathbf{p} + \frac{1}{3}\mathbf{t}$ or $\frac{1}{3}(-\mathbf{p} + \mathbf{t})$ final answer | 2 | **M1** for $\overrightarrow{PT} = -\mathbf{p} + \mathbf{t}$ or $\overrightarrow{TP} = -\mathbf{t} + \mathbf{p}$ |
| 22(a)(ii) | $\frac{2}{3}\mathbf{p} - \frac{1}{3}\mathbf{t}$ or $\frac{1}{3}(2\mathbf{p} - \mathbf{t})$ final answer | 2 | **M1** for a correct route |
| 22(b) | $\overrightarrow{OM} = \frac{2}{3}\mathbf{t} - \frac{2}{3}\mathbf{t} + \frac{4}{3}\mathbf{p}$ oe | M1 | |
| | $\overrightarrow{OM} = \frac{4}{3}\mathbf{p}$ and<br>$\overrightarrow{OM}$ is a multiple of $\overrightarrow{OP}$ oe | A1 | |
| 23 | 3.6 | 3 | **M1** for $\frac{\textit{their}\text{ distance}}{\textit{their}\text{ time}}$<br>**B1** for 7.95 or 7950 or 132.5 |

© Cambridge University Press & Assessment 2022

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 24 | (–2.79, –4.58)<br>(1.79, 4.58) | **6** | **B5** for (–2.791... , –4.583 to –4.582)<br>**and** (1.791..., 4.582 to 4.583)<br>or **B4** for –2.79 or –2.791... **and** 1.79 or 1.791...<br><br>OR<br><br>**M1** for $2x + 1 = x^2 + 3x - 4$ or better<br><br>**M2** for $\frac{-1 \pm \sqrt{1^2 - 4(1)(-5)}}{2(1)}$<br>**FT** *their* quadratic not $x^2 + 3x - 4$<br>or **M1** for $\frac{-1 \pm \sqrt{p}}{2(1)}$ or $\sqrt{1^2 - 4(1)(-5)}$<br>**FT** *their* quadratic not $x^2 + 3x - 4$ |

© Cambridge University Press & Assessment 2022